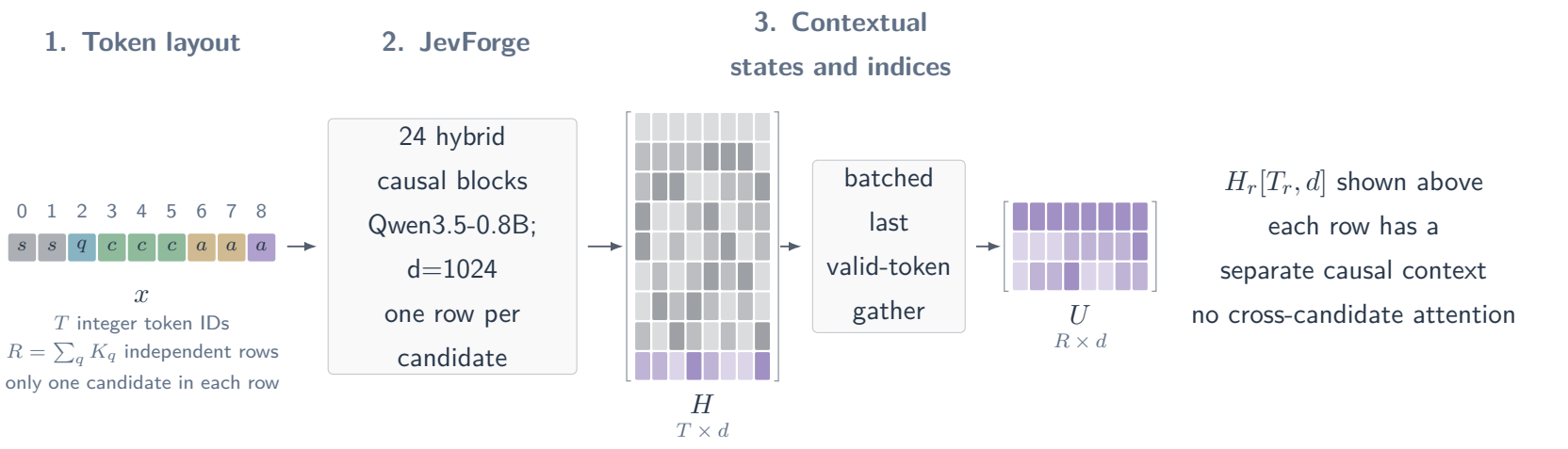
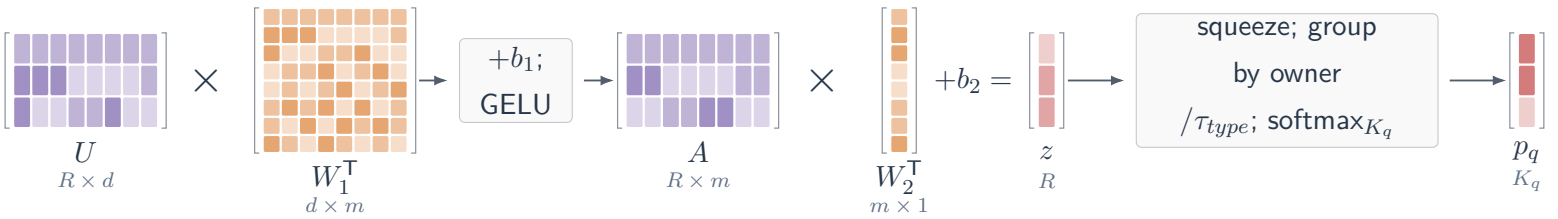
$$U_r = H_r[t_r, :],\ t_r = \sum_t A^{pad}_{r,t} - 1, \quad A = \mathrm{GELU}(UW_1^\mathsf{T} + b_1), \quad s = \mathrm{squeeze}(AW_2^\mathsf{T} + b_2)$$





Axes $d = 1024$, $m = 1024$; R: independent candidate rows, not questions. The illustration shows one question with $R = K = 3$. Equal d and m edges reflect this head's equal width.

Objects Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes.

Mechanism Raw pair scores are independent of other candidate texts. Owner grouping and candidate softmax couple probabilities; no shared-prefix tree kernel is implemented.

Pinned release: candidate CE + 0.5 Brier; 12 head-only steps, then full-model tuning. The separate RLCD implementation is not this checkpoint history.